
UNDERGRADUATE THESIS REPORT 1

The Mass Fallback Rate and Light Intensity Curve generated by a Tidal Disruption Event on a Constant Density Star

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1 Introduction

Following the work of Oña et al.[1], we analyze the effect of Tidal Disruption Events (TDEs) on the mass fallback rate of the disrupted stellar material $\dot{M}$, as well as the characteristic luminosity curve L . In this work, we hope to recover the canonical result $\dot{M} \propto T^{-5/3}$ fallback rate established by Rees[2] and Phinney[3], as well as some interesting results under the assumption that the star is homogeneous, i.e., constant density.

A TDE occurs when a star strays too close to a supermassive black hole and is torn apart by tidal forces, flinging its debris in to a range of orbits. The general procedure to determine the mass fallback rate rests on the chain rule

$$\frac{dM}{dT} = \frac{dM}{dE} \frac{dE}{dT}, \quad (1.1)$$

which factorizes the problem into two parts; a *stellar part* and an *black hole part*. The stellar part determines how the star's mass is distributed across binding energies, which depends on the star's internal density profile. This density profile is usually treated as polytropic, however in this work we take it to be isotropic and homogeneous. The black hole part determines how each debris parcel's orbital energy return over time, via Kepler's third law. Multiplying the two returns the mass fallback rate, and since a fraction of the infalling debris' energy is radiated as it circularizes and accretes, this in turn produces an observable luminosity curve

$$L = \eta \dot{M} c^2 \quad (1.2)$$

where η is the radiative efficiency of the blackhole, i.e., the fraction of the infalling rest-mass energy that escapes as radiation, $\dot{M} c^2$ being the rest-mass energy flux of the fallback debris. In this work, we take a different but similar approach to arrive with $\dot{M}$ and L .

2 Disruption Dynamics

As a starting point, we consider a singular Schwarzschild blackhole with mass M_{BH} . We don't consider other effects such as galactic gravitational potential or other extrinsic effects as of yet. The entire derivation will be purely Newtonian in nature. This is the simplest possible case, being a nontorating, spherically symmetric system. At far distances r , the relativistic effects are negligible, and the entire system can be approximated by Newtonian dynamics. In this case, the black hole can be approximated as a point mass. This has a typical radial inverse gravitational potential.

$$\Phi_{\text{BH}} = -\frac{GM_{\text{BH}}}{r} \quad (2.1)$$

From here on, we just call it Φ . Now, we specify the encounter. We consider a star, of mass M_* , entering at a typical assumed parabolic trajectory, i.e., the characteristic orbital energy $E = 0$. We

specify it so that the trajectory touches the tidal disruption radius, R_t , precisely. That is, it has a penetration factor δ of 1.

$$\delta = \frac{R_t}{R_p} \quad (2.2)$$

where R_p is the periapsis (called peribothron for blackholes). The system is sketched in Figure 1

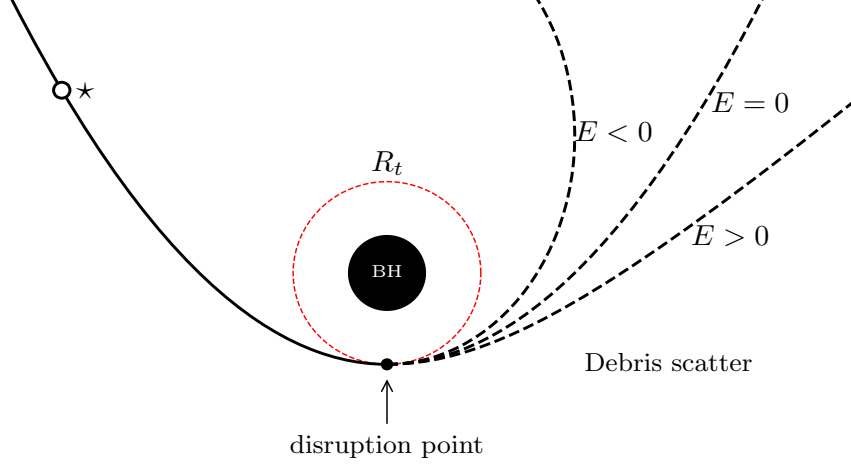


Figure 1: A star approaching a black hole at R_t

Tidal disruption occurs when the tidal force exceeds the star's self gravity, i.e.,

$$\frac{d\Phi(r - R_\star)}{dr} - \frac{d\Phi(r + R_\star)}{dr} = \frac{GM_\star}{R_\star^2} \quad (2.3)$$

Solving for r gives R_t . For a typical Newtonian tidal radius, and when $R_\star \ll r$, this can be approximated by

$$R_t = R_\star \left(\frac{4M_{\text{BH}}}{M_\star} \right)^{1/3} \quad (2.4)$$

This is classically known as the Roche Limit (up to a factor of unity). The precise derivation is found in the [appendices](#).

Before the disruption, the center of mass of the star has energy

$$E_\star = 0$$

However, upon disruption, different parts of the star, depending on the distance from the blackhole, are imparted with different energies. This is expressed as

$$\Delta E \simeq \left(\frac{d\Phi}{dr} \right)_{R_p} R_\star \quad (2.5)$$

For an initial approximation, $d\Phi/dr$ is

$$\frac{d\Phi}{dr} = \frac{GM_{\text{BH}}}{r^2}$$

therefore,

$$\Delta E \simeq \frac{GM_{\text{BH}}}{R_p^2} R_\star$$

After disruption, parts of the star acquire orbital energy $E \geq 0$, which means it is unbound, and $E < 0$, which means it is bound. For this, we assume the “frozen-in” approximation, where we assume that each part of the star instantaneously acquire the orbital energy at the point of disruption, and evolves ballistically over time.

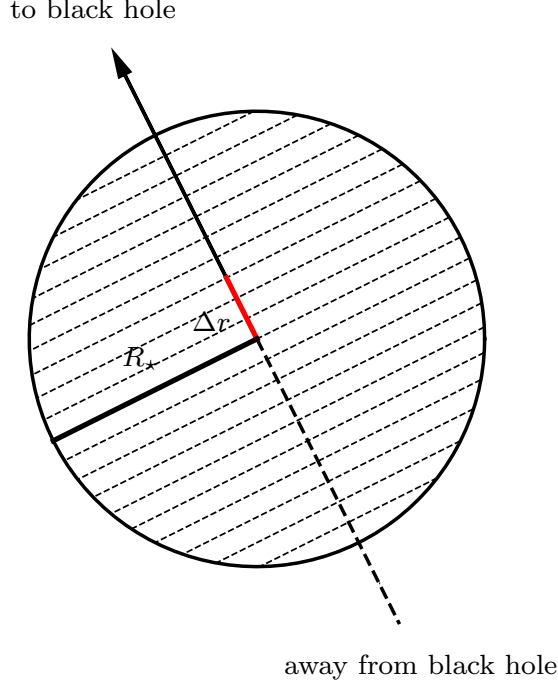


Figure 2: “Slicing” the star to get the energy imparted

Now, we imagine “slicing” the star in the direction perpendicular to the direction of the blackhole. Let Δr be the distance of the slice from the center of the star, where $\Delta r > 0$ is towards the blackhole, and $\Delta r < 0$ is away (Fig. 2).

From Lodato[4], we adapt a dimensionless parameter ξ such that

$$\xi = \frac{\Delta r}{R_*} \quad (2.6)$$

The orbital energy of the star is then approximately linear in its profile

$$E(\Delta r) \simeq -\frac{GM_{\text{BH}}}{R_p^2} \Delta r$$

We can then express E in terms of ξ .

$$E\xi = -\Delta E \quad \implies \quad E(\xi) = -\Delta E\xi$$

Equivalently,

$$\xi = -\frac{E}{\Delta E}$$

The near side of the star has $\xi \geq 0$, thus $E \leq 0$ (bound), and the far side has $\xi < 0$, therefore $E > 0$ (unbound).

So for the accretion curve, we only really care about

$$0 < \xi \leq 1$$

3 Stellar Structure

Now, we need to know how much mass has each energy, i.e., we need to find $\frac{dM}{dE}$.

Because $E = -\Delta E \xi$, then $\frac{dE}{d\xi} = -\Delta E$. Therefore,

$$\frac{dM}{dE} = -\frac{1}{\Delta E} \frac{dM}{d\xi}$$

To find $dM/d\xi$, we need to consider how much mass is in each slice of the star. The mass element is

$$dM = 2\pi d(\Delta r) \int_{\Delta r}^{R_\star} \rho(r) r dr \quad (3.1)$$

where r is the radius of each slice from the center of the star (derivation at the Appendices). From (3.1), we can find the mass per each slice.

$$\frac{dM}{d\Delta r} = 2\pi \int_{\Delta r}^{R_\star} \rho(r) r dr$$

For a dimensionless form, we can consider that since $\xi = \frac{\Delta r}{R_\star}$, we have $r = R_\star \xi'$, thus

$$\frac{dM}{d\xi} = 2\pi R_\star^3 \int_{\xi}^1 \rho(R_\star \xi') \xi' d\xi'$$

From the work of Oña et al, they consider a polytropic star, where the density profile is spherically symmetric, $\rho(r)$, and can be solved by the Lane-Emden equation[5]

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n \quad (3.2)$$

Where θ is a dimensionless density, and ξ is a dimensionless radius. In the work, they use a polytropic index of $n = 3$, which corresponds to solar-type main sequence stars. However, for the case of the current work, we consider a constant density star, i.e., the polytropic index is $n = 0$. The Lane-Emden equation then reduces to

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -1.$$

With the boundary conditions $\theta(0) = 1$ and $\theta' = 0$, the solution is simply

$$\theta(\xi) = 1 - \frac{\xi^2}{6}$$

For $\theta = 0$, we find the radius of the star. It turns out to be $R_\star = a\sqrt{6}$, where a is the radial scale length. That isn't really the point here, the point is that we now find that the star has a density profile of

$$\rho(r) = \begin{cases} \rho_0, & 0 \leq r \leq R_\star \\ 0, & r > R_\star \end{cases} \quad (3.3)$$

where ρ_0 is the average density of the star. This density profile makes it possible to solve the stellar part of the procedure to be completely analytic. That is,

$$\begin{aligned} \frac{dM}{d\Delta r} &= 2\pi \int_{\Delta r}^{R_\star} \rho_0 r dr \\ &= 2\pi \rho_0 \int_{\Delta r}^{R_\star} r dr \\ &= 2\pi \rho_0 \left[\frac{r^2}{2} \right]_{\Delta r}^{R_\star} \\ &= 2\pi \rho_0 \left(\frac{R_\star^2}{2} - \frac{\Delta r^2}{2} \right) \\ &= \pi \rho_0 (R_\star^2 - \Delta r^2) \end{aligned} \quad (3.4)$$

Equivalently, in dimensionless form,

$$\begin{aligned}
\frac{dM}{d\xi} &= 2\pi\rho_0 R_\star^3 \int_\xi^1 \xi' d\xi' \\
&= 2\pi\rho_0 R_\star^3 \left. \frac{\xi'^2}{2} \right|_\xi^1 \\
&= 2\pi\rho_0 R_\star^3 \left(\frac{1}{2} - \frac{\xi^2}{2} \right) \\
&= \pi\rho_0 R_\star^3 (1 - \xi^2)
\end{aligned} \tag{3.5}$$

4 Fallback Dynamics

For the black hole orbital dynamics, we proceed with the following. Each band of debris has original specific orbital energy $E = 0$. However, post-disruption, they each receive different energies. For a keplerian orbit, the specific orbital energy is

$$E = -\frac{GM_{\text{BH}}}{2a} \tag{4.1}$$

where a is the semi-major axis of the orbit. We can solve for it, and we find that

$$a = -\frac{GM_{\text{BH}}}{2E}$$

Kepler's third law gives the period of the orbit

$$T = 2\pi\sqrt{\frac{a^3}{GM_{\text{BH}}}}$$

Substituting a , we have

$$\begin{aligned}
T &= 2\pi\sqrt{\frac{1}{GM_{\text{BH}}} \left(-\frac{GM_{\text{BH}}}{2E} \right)^3} \\
&= 2\pi GM_{\text{BH}} (-2E)^{-3/2}
\end{aligned}$$

Since $E = -\Delta E \xi$, we get

$$T(\xi) = 2\pi GM_{\text{BH}} (-2\Delta E \xi)^{-3/2}$$

We want the amount of material returning to R_p at a particular time T after the disruption, so we invert the function.

$$\xi(T) = \frac{1}{2\Delta E} \left(\frac{2\pi GM_{\text{BH}}}{T} \right)^{2/3} \tag{4.2}$$

At time T , the returning debris are those which have orbital period T . Therefore, we have

$$\frac{dM}{dT} = \frac{dM}{d\xi} \frac{d\xi}{dT} \tag{4.3}$$

Notice that this is the same form as (1.1), but we swapped the energy E to the dimensionless parameter ξ , i.e.,

$$\frac{dM}{dT} = \left(-\frac{1}{\Delta E} \frac{dM}{d\xi} \right) \left(-\Delta E \frac{d\xi}{dT} \right)$$

Moving forward, we find that

$$\frac{d\xi}{dT} = -\frac{(2\pi GM_{\text{BH}})^{2/3}}{3\Delta E} T^{-5/3} \tag{4.4}$$

Multiplying this with (3.5), we find that

$$\frac{dM}{d\xi} \frac{d\xi}{dT} = -\pi\rho_0 R_\star^3 (1-\xi^2) \frac{(2\pi GM_{\text{BH}})^{2/3}}{3\Delta E} T^{-5/3} \quad (4.5)$$

Since $\rho_0 = \frac{3M_\star}{4\pi R_\star^3}$, then $\frac{\pi\rho_0 R_\star^3}{3} = \frac{M_\star}{4}$. Hence

$$\frac{dM}{dT} = \frac{M_\star (2\pi GM_{\text{BH}})^{2/3}}{4\Delta E} (1-\xi^2) T^{-5/3}$$

Expanding $\xi(T)$ from (4.2), we have

$$\dot{M} = \frac{M_\star (2\pi GM_{\text{BH}})^{2/3}}{4\Delta E} \left(1 - \frac{1}{4\Delta E^2} \left(\frac{2\pi GM_{\text{BH}}}{T} \right)^{4/3} \right) T^{-5/3}$$

Simplifying this, we have the explicit expression for $\dot{M}$

$$\dot{M} = \frac{M_\star (2\pi GM_{\text{BH}})^{2/3}}{4\Delta E} T^{-5/3} - \frac{M_\star (2\pi GM_{\text{BH}})^2}{16\Delta E^3} T^{-3} \quad (4.6)$$

We can further expand ΔE , but this is simply a constant independent of time, so I elected not to do so. Hence, we have derived the mass fallback rate for a constant density star (for the appropriate assumptions of approach specific orbital energy $E_\star = 0$ and penetration factor $\delta = 1$). However, at this point we indeed recover the canonical $\dot{M} \propto T^{-5/3}$ result, as well as a following T^{-3} term.

To find the light curve generated by this fallback rate, we simply multiply it by ηc^2 . The value of η is approximately $\eta \approx 0.1$ for a Schwarzschild black hole [6]. The expression for the light curve is then

$$L = \eta c^2 \left(\frac{M_\star (2\pi GM_{\text{BH}})^{2/3}}{4\Delta E} T^{-5/3} - \frac{M_\star (2\pi GM_{\text{BH}})^2}{16\Delta E^3} T^{-3} \right) \quad (4.7)$$

5 Fallback rate and Light curve plots

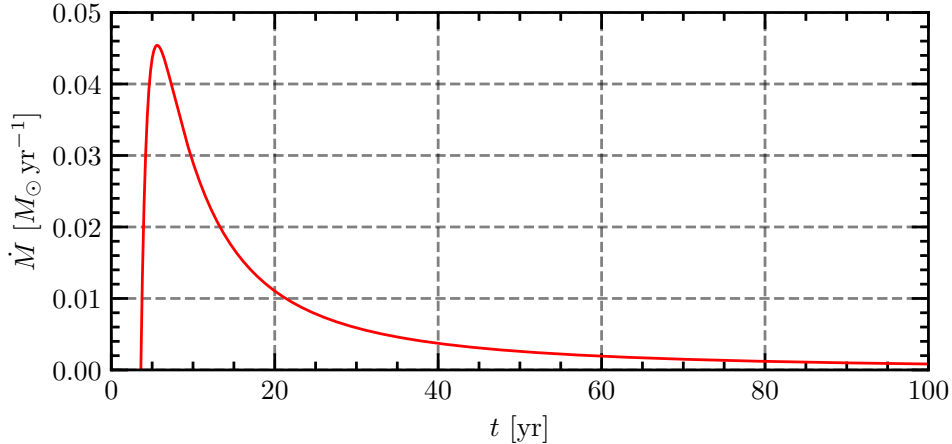


Figure 3: Mass fallback rate of a solar-mass star disrupted by a $10^7 M_\odot$ black hole.

Figure 3 shows the mass fallback rate of a $1 M_\odot$ star disrupted by a supermassive black hole with mass $10^7 M_\odot$, assuming a parabolic center-of-mass orbit with $E_\star = 0$ and a penetration factor of $\delta = 1$. The time t denotes the elapsed time since the disruption, measured in years. At early times, approximately 0 – 5 years after disruption, the fallback rate initially increases as the negative T^{-3} term dominates the expression for $\dot{M}$. The fallback rate subsequently reaches a maximum, after which it decreases as the $T^{-5/3}$ term becomes dominant at late times. Thus, the long-time behavior approaches the canonical power-law dependence $\dot{M} \propto T^{-5/3}$.

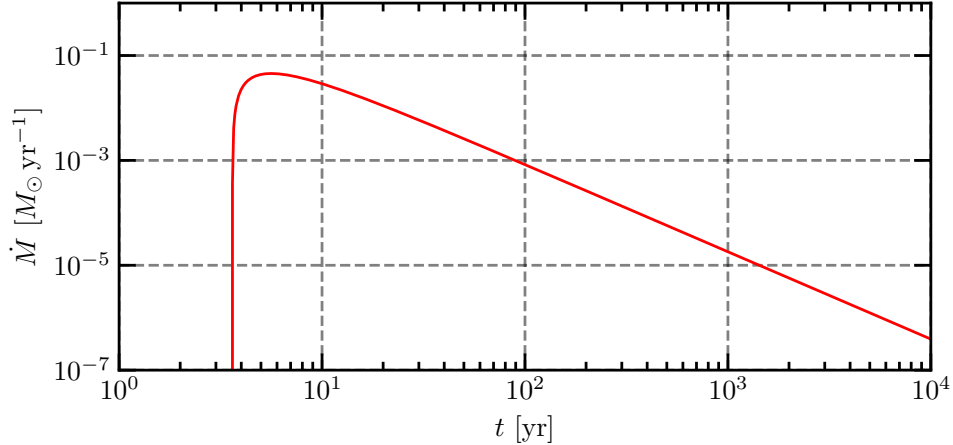


Figure 4: Mass fallback rate of the same tidal disruption event as in Figure 3, shown on a log-log scale.

Figure 4 shows the same event plotted on a log-log scale. The late-time fallback follows an approximately linear trend on the log-log plot, corresponding to the power-law dependence $\dot{M} \propto T^{-5/3}$. This asymptotic behavior becomes increasingly apparent at longer times as the T^{-3} contribution becomes negligible compared with the $T^{-5/3}$ contribution.

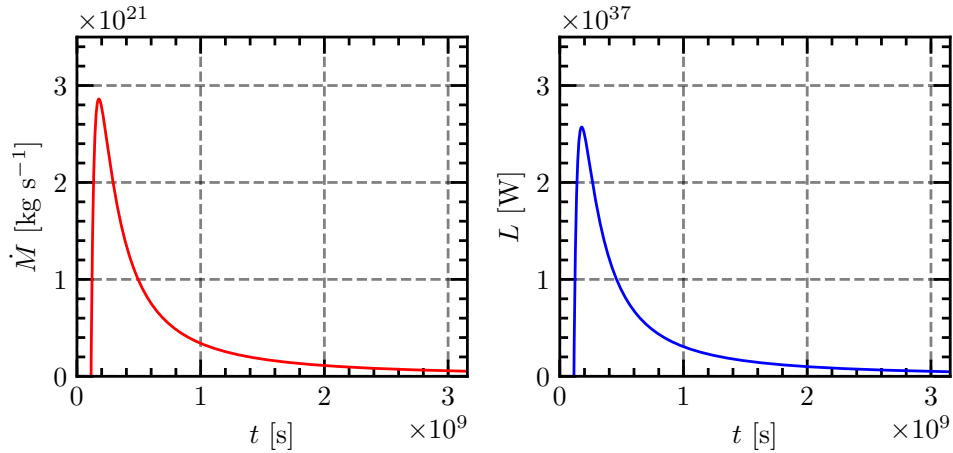


Figure 5: Left: Accretion rate in SI units. Right: Corresponding luminosity curve generated from the fallback rate.

Figure 5 shows the same event expressed in SI units. The left panel gives the mass fallback rate in kg s^{-1} as a function of time in seconds. The right panel shows the corresponding luminosity curve in watts, obtained using the accretion luminosity prescription $L = \eta \dot{M} c^2$, with a constant radiative efficiency $\eta = 0.1$. Since η and c are constant, the luminosity is directly proportional to the fallback rate, $L \propto \dot{M}$. Consequently, the luminosity curve has the same temporal dependence and overall shape as the mass fallback rate.

Figure 6 shows the mass fallback rate of a $1M_\odot$ star for black holes with masses ranging from $10^3 M_\odot$ to $10^8 M_\odot$. As in the previous calculations, the center-of-mass orbit is assumed to be parabolic, with $E_\star = 0$ and $\delta = 1$. The peak fallback time varies with black-hole mass because the characteristic return time of the debris depends on the energy spread induced by the tidal interaction. More massive black holes produce a different characteristic fallback timescale, shifting the location of the peak in time. Despite this variation in the peak time and fallback rate, the late-time behavior of all cases approaches the same $T^{-5/3}$ power-law dependence predicted by the model.

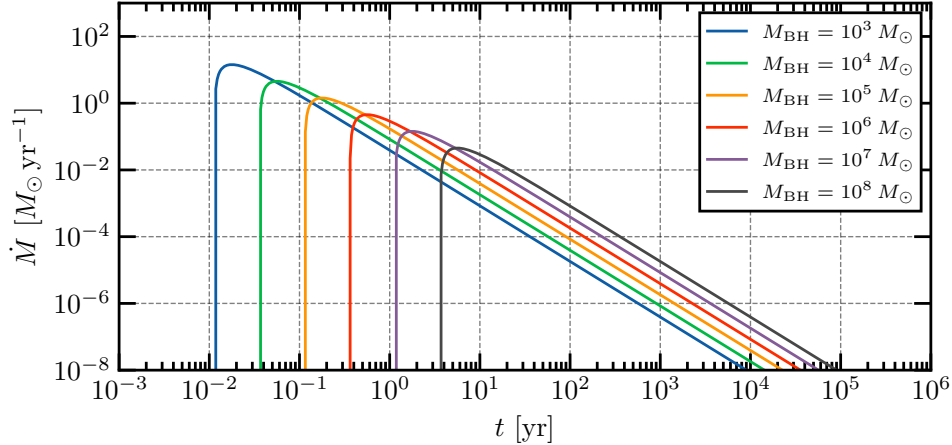


Figure 6: Mass fallback rates for a $1M_\odot$ star disrupted by black holes with masses ranging from 10^3 to $10^8 M_\odot$

Overall, we see from visual inspection that the mass fallback rate obtained under the constant-density assumption closely resembles that obtained under a polytropic stellar model shown in the work of Oña. In particular, both models exhibit an initial rise toward a peak fallback rate followed by a power-law decline at late times. This suggests that the constant-density model may provide a reasonable analytic approximation for capturing the general temporal behavior of the fallback rate.

However, the apparent agreement between the two models is based only on the comparison of the resulting curves and does not establish quantitative equivalence. The internal density profile of a realistic star differs substantially from the constant-density assumption, particularly toward the stellar center, and these differences can affect the distribution of debris orbital energies and consequently the detailed shape of the fallback curve. A more rigorous comparison would therefore require numerical calculations using realistic stellar density profiles, such as polytropic or numerically evolved stellar models. Such simulations would allow the magnitude of the differences between the constant-density and polytropic results to be quantified and would determine the range over which the constant-density approximation remains valid.

Appendices

Derivation of the tidal radius

From the potential $\Psi = -\frac{GM_{\text{BH}}}{r}$, we have

$$\frac{d\Psi(r - R_\star)}{dr} = \frac{GM_{\text{BH}}}{(r - R_\star)^2}$$

and

$$\frac{d\Psi(r + R_\star)}{dr} = \frac{GM_{\text{BH}}}{(r + R_\star)^2}$$

Therefore

$$\frac{GM_{\text{BH}}}{(r - R_\star)^2} - \frac{GM_{\text{BH}}}{(r + R_\star)^2} = \frac{GM_\star}{R_\star^2}$$

We cancel G ,

$$M_{\text{BH}} \left(\frac{1}{(r - R_\star)^2} - \frac{1}{(r + R_\star)^2} \right) = \frac{M_\star}{R_\star^2}$$

Evaluating the difference, we have

$$\frac{1}{(r - R_\star)^2} - \frac{1}{(r + R_\star)^2} = \frac{4rR_\star}{(r^2 - R_\star^2)^2}$$

Hence

$$\frac{4rR_\star}{(r^2 - R_\star^2)^2} = \frac{M_\star}{R_\star^2}$$

Therefore

$$(r^2 - R_\star^2)^2 = 4 \frac{M_{\text{BH}}}{M_\star} r R_\star^3$$

For $r \gg R_\star$, since $(r^2 - R_\star^2)^2 \simeq r^4$, we obtain

$$r^4 \simeq 4 \frac{M_{\text{BH}}}{M_\star} r R_\star^3$$

so

$$r^3 \simeq 4 \frac{M_{\text{BH}}}{M_\star} R_\star^3$$

Thus

$$R_t \simeq \left(4 \frac{M_{\text{BH}}}{M_\star} \right)^{1/3} R_\star$$

Derivation of the mass element dM

For the star slices, we take the cylindrical slice at a distance Δr from the center of the star. We introduce a variable x , which is the radius of the cylinder at Δr . At the outer edge of the slice, x is related to the spherical radius r by

$$x^2 = R_\star^2 - \Delta r^2$$

To get the mass of the slice, and since the density is not uniform but is instead spherically symmetric, we approximate it as a series of concentric rings with radius x , and height $d\Delta r$. We start from the center of the slice at $x = 0$, and at end at the edge of the star, at $x = \sqrt{R_\star^2 - (\Delta r)^2}$. The mass of the integrated cylinder is therefore

$$dM = 2\pi d\Delta r \int_0^{\sqrt{R_\star^2 - (\Delta r)^2}} \rho'(x) x dx$$

where ρ' is the density expressed as a function of the cylindrical radius x . Since the slice is located at a distance Δr from the center, the cylindrical x and the spherical radius r are related by

$$x^2 = r^2 - (\Delta r)^2$$

Thus, the density function $\rho'(x)$ is related to the spherical density function $\rho(r)$ by

$$\rho'(x) = \rho\left(\sqrt{x^2 + (\Delta r)^2}\right)$$

To express the integral in terms of the spherical radius r , we differentiate $r^2 = x^2 + (\Delta r)^2$, where Δr is fixed for a given slice. We find that

$$2r \, dr = 2x \, dx$$

or

$$x \, dx = r \, dr$$

At the center of the slice, $x = 0$, therefore $r = \Delta r$, and at the edge, $x = \sqrt{R_\star^2 - \Delta r^2}$, and hence, $r = R_\star$. We now substitute $r \, dr$ in the integral and change the limits. We can also substitute in $\rho(r)$, the spherical density function. We now have the mass element of the slice in terms of the spherical radius

$$dM = 2\pi \, d\Delta r \int_{\Delta r}^{R_\star} r \rho(r) \, dr$$

References

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